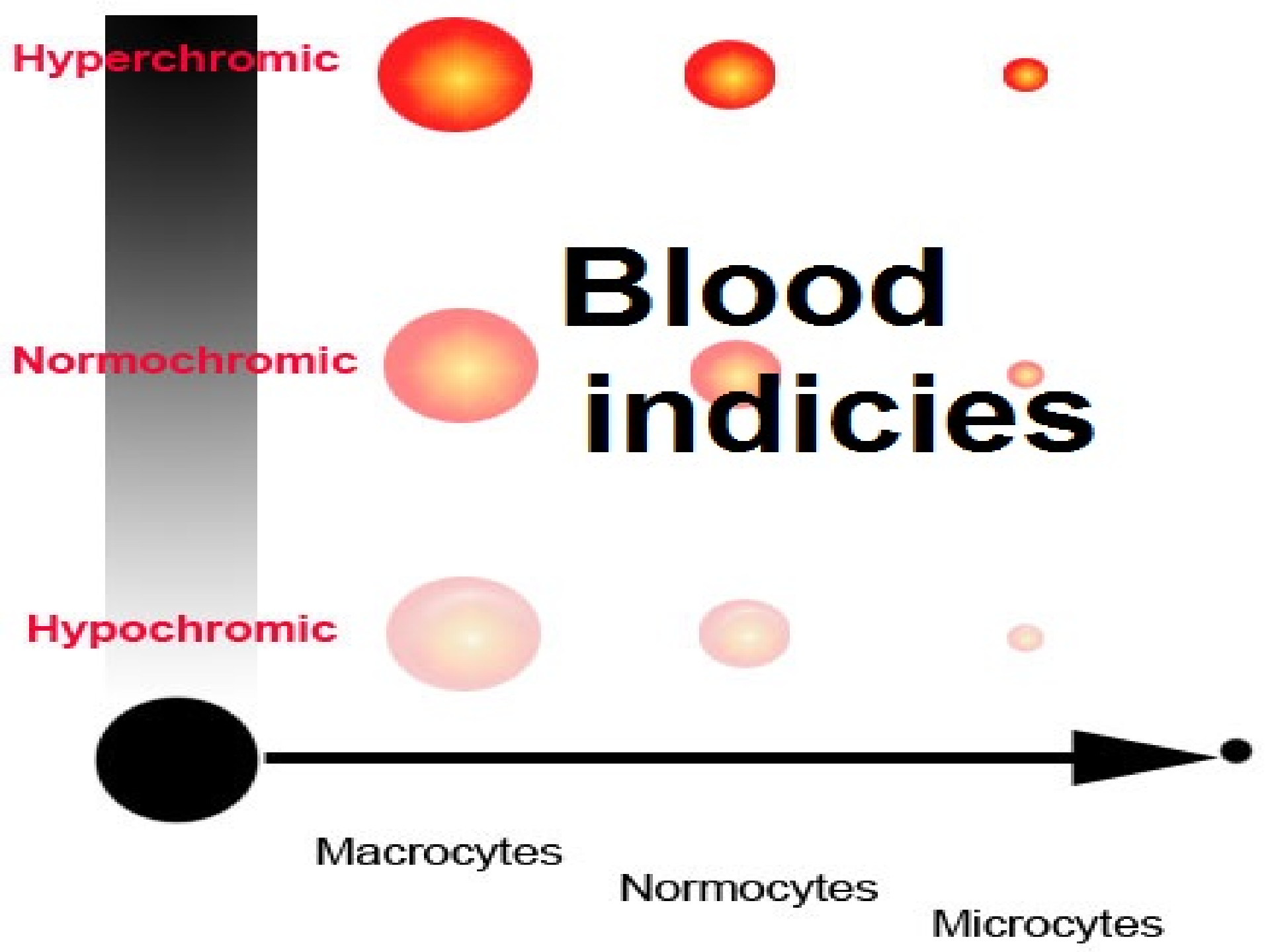
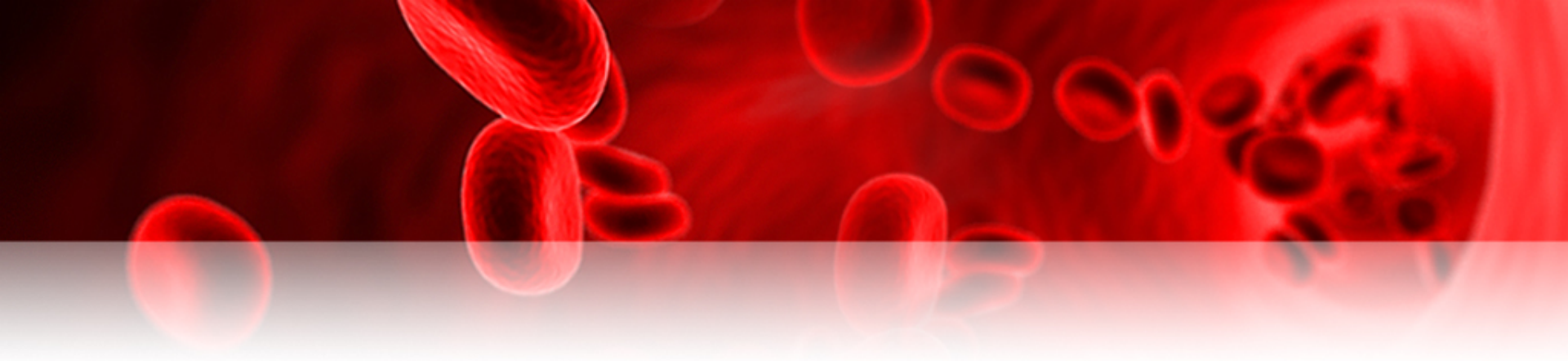

Hyperchromic
Normochromic
Hypochromic
Blood indicies
Macrocytes
Normocytes
Microcytes

BLOOD INDICES

Definition :

They are calculations that provide information about haemoglobin content and size of RBCs which aid in diagnosis and classification of anemia.



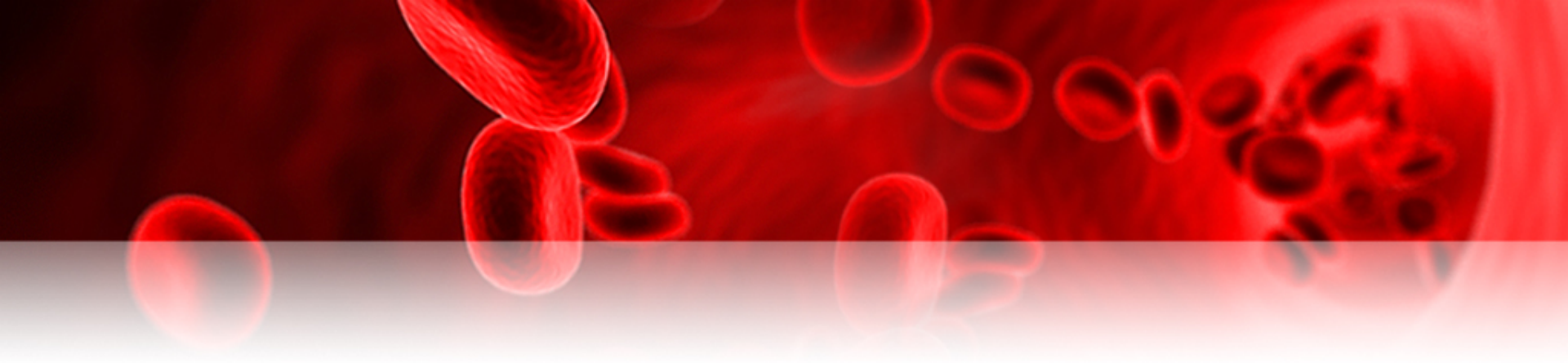
- **Anemia** is a decrease in RBCs number and hemoglobin concentration resulting in a reduced ability for the blood to carry oxygen to body tissues.

- **Normal RBCs count**

  Male: 5-6 million/µL
  Female: 4-5 million/µL

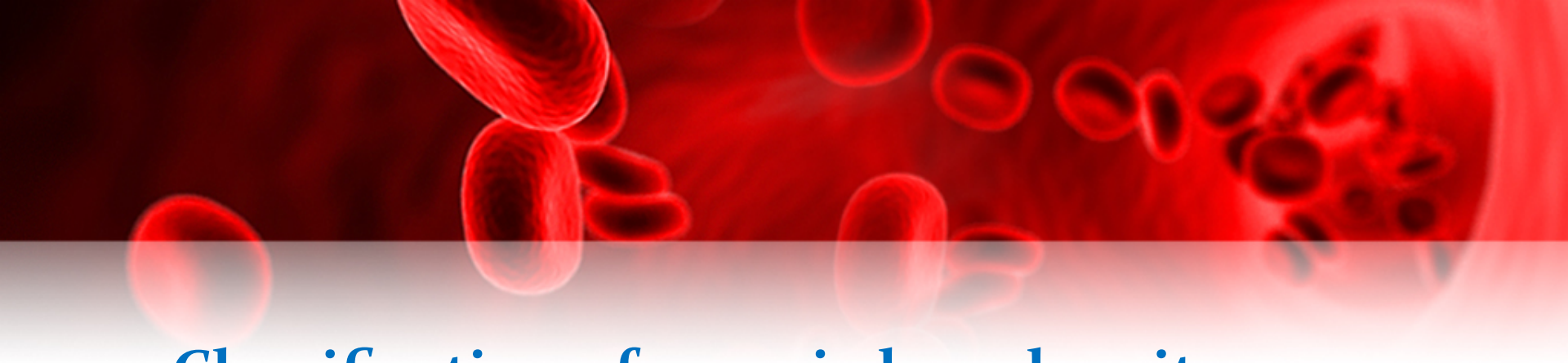
- **Normal hemoglobin concentration**

  Male: 14-16 g/dL
  Female: 12-15 g/dL



Classification of Anemia

- On the basis of the cause
  - Decreased production
  - Increased loss
- On the basis of morphology
  - Microcytic
  - Normocytic
  - Macrocytic



Classification of anemia based on its cause

Decreased production

Deficiency anemia

- Iron deficiency anemia
- Vitamin B12 deficiency anemia
- Folic acid deficiency anemia

Aplastic anemia

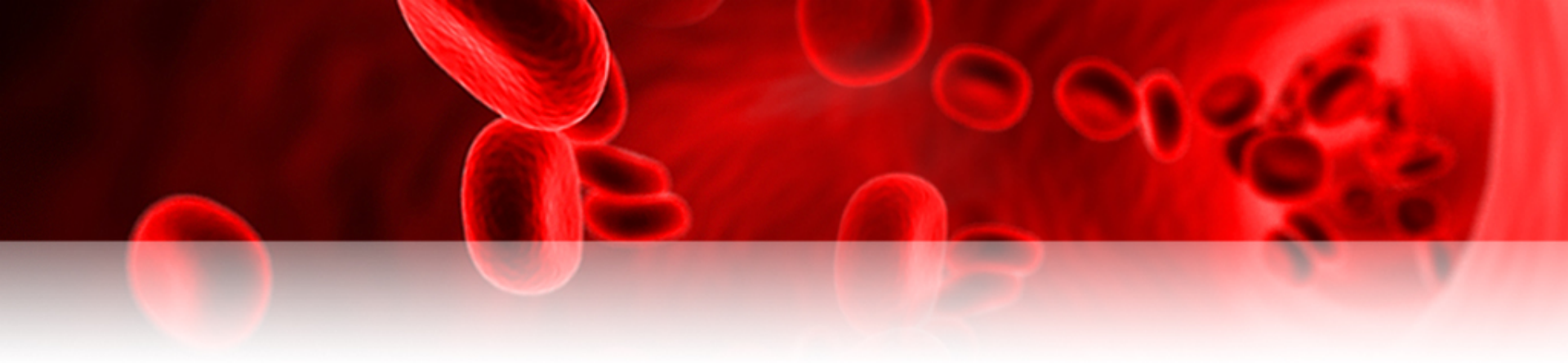
Anemia of chronic disease

Increased loss

Hemolytic anemia

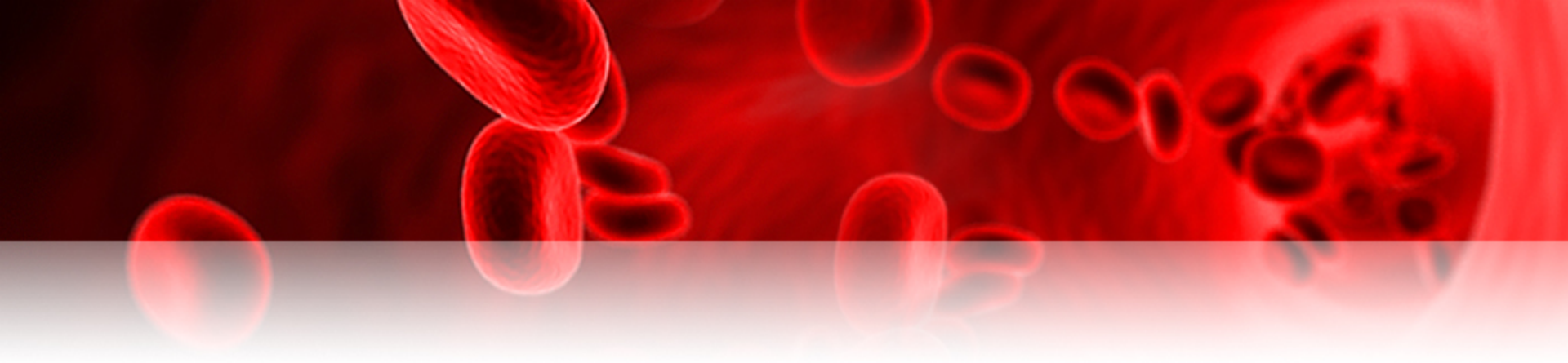
- Hemoglobinopathy → Sickle cell anemia and thalassemia
- Enzyme deficiency → G6PD deficiency
- Membrane defect → Hereditary sherocytosis

Hemorrhagic anemia



Classification of anemia based on its morphology

Microcytic (< 80 fL)	Normocytic (80-95 fL)	Macrocytic (> 100 fL)
• Iron deficiency anemia • Thalassemia • Anemia due to chronic bleeding	• Aplastic anemia • Hemolytic anemia • Acute hemorrhagic anemia	• Vitamin B12 deficiency anemia • Folic acid deficiency anemia



How to Diagnose Anemia

- **Complete blood count (CBC)**

Test description	Observed value	Unit	Reference range
Erythrocytes			
Total count	4.21	$\times 10^6$/μL	3.8–5.4
Hemoglobin	9.6	g/dL	10.5–14.0
PCV (hematocrit)	30.1	%	32–42
MCV	71.5	fL	72–88
MCH	22.8	pg	24–30
MCHC	31.9	g/dL	32–36
Leucocytes			
Total leucocyte count	11,700	%	4400–11,300
Neutrophils	31	%	45–74
Lymphocytes	66	%	22–50
Basophils	00	%	0–1
Eosinophils	02	%	0–4
Monocytes	01	%	1–8
Platelets			
Total count	840	$\times 10^3$/μL	10–400
MPV	7.7	fL	8–12
PDW	8.8	fL	9–14

BLOOD INDICES

They include:

1. Mean corpuscular volume **(MCV)**
2. Mean corpuscular haemoglobin **(MCH)**
3. Mean corpuscular haemoglobin concentration **(MCHC)**

Mean corpuscular volume (MCV)

It is the average size of a RBC.

It is calculated by dividing HV by RBCs count.

$$\text{MCV} = \frac{\text{H.V} \times 10}{\text{RBCs count in millions /µL}}$$

Normally : **80 - 95 fL (µm³)**

Microcyte **< 80 fL** e.g. iron deficiency anemia.

Macrocyte **> 95 fL** e.g. vitamin B12 deficiency anemia.

NB. Femtoliter = 10^{-15} liter

Mean corpuscular hemoglobin (MCH)

It is the average amount of haemoglobin per RBC.

It is calculated by dividing the haemoglobin by RBCs count

$$\text{MCH} = \frac{\text{Hb content (g/dL)} \times 10}{\text{RBCs count in millions/µL}}$$

Normally : **27: 33 pg**

NB . Picogram = 10^{-12} g

Mean corpuscular haemoglobin concentration (MCHC)

It is the average Hb concentration in a single RBC.

It Is calculated by dividing the haemoglobin by the hematocrit.

$$\text{MCHC} = \frac{\text{Hb content (g/dL)} \times 100}{\text{hematocrit}}$$

Normally : **33 %** → Normchromic

< 33 % → Hypochromic anemia e.g. iron deficiency anemia.

MCHC is not changed in vitamin B12 deficiency.

BLOOD INDICES

$$MCV = \frac{H.V}{\text{RBCs count in millions } /\mu L} \times 10$$

$$MCH = \frac{Hb}{\text{RBCs count in millions } /\mu L} \times 10$$

$$MCHC = \frac{Hb}{H.V} \times 100$$

Case study

A 55 year-old male, smoker complained of easy fatigability, palpitation, dyspnea on exertion and occasional epigastric pain, he gave history of H. pylori infection 3 years ago for which he received medical treatment but he didn't follow up. O/E there was pallor, pulse 95 bpm, ABP 130/90, complete blood count revealed the following: hemoglobin 6 g%, PCV 24% and the RBCs count was 4000,000/µl.

1- Calculate the blood indices of this patient & comment on it.

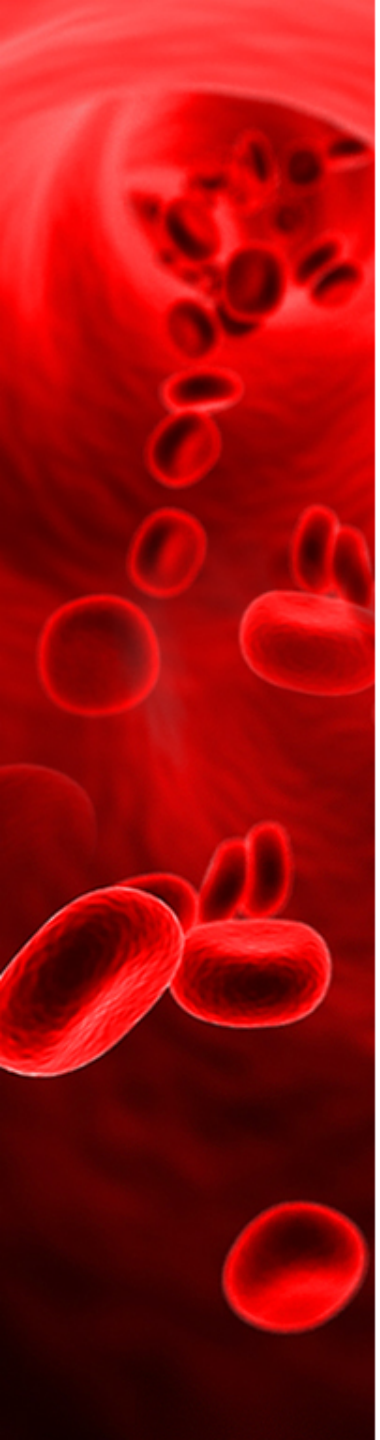
2- Diagnose the type of anemia.

$$\text{1- MCV} = \frac{\text{HV}}{\text{RBCs count in millions /mm}^3} \times 10$$

$$= \frac{24}{4} \times 10$$

$$= 60 \text{ fl}$$

Normally :
80 - 95 fl



$$2\text{- MCH} = \frac{\text{Hb}}{\text{RBCs count in millions /mm}^3} \times 10$$

$$= \frac{6}{4} \times 10$$

$$= 15 \text{ Pg}$$

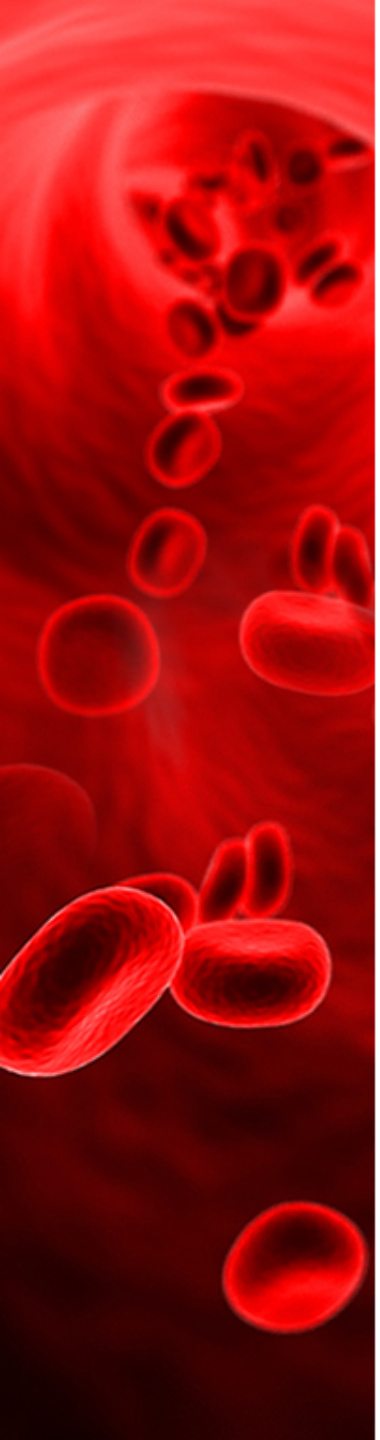
Normally : 27 - 33 pg

$$2\text{- MCHC} = \frac{\text{Hb}}{\text{Hv}} \times 100$$

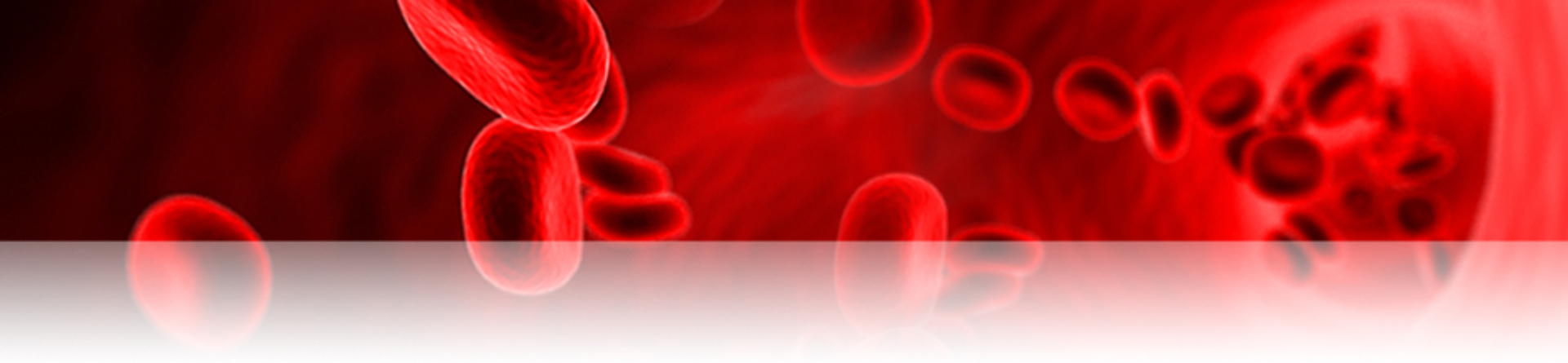
$$= \frac{6}{24} \times 100$$

$$= 25\ \%$$

Normally : 33 %



- So, this patient has **Microcytic Hypochromic Anemia** and he has **Iron deficiency anemia.**



- A 15 year-old boy on a vegetarian diet, his blood picture showed the following: Hb 10 g/100 ml, RBCs 2,000,000/ mm^3, MCV 106 fL and MCH 34 pg.

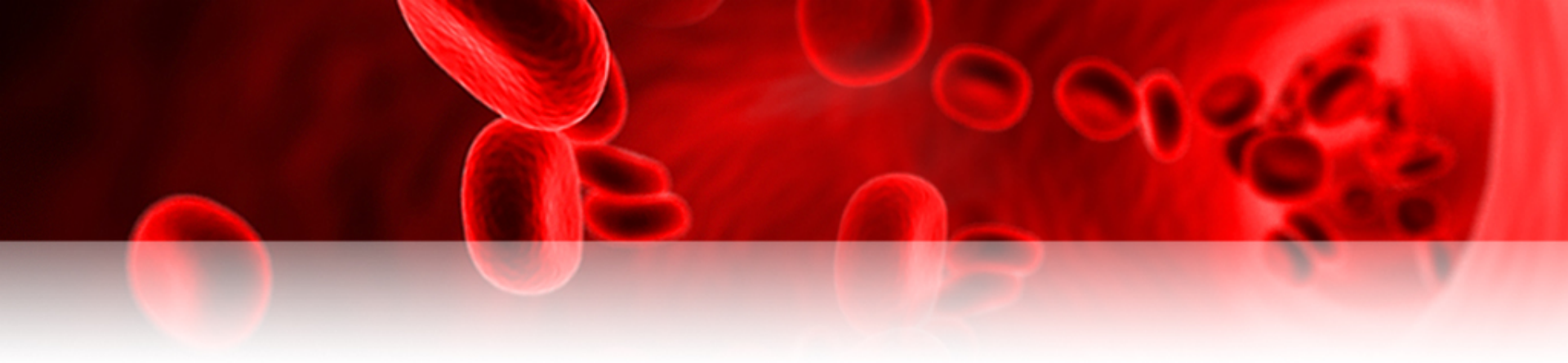
Choose the correct answer:

A- The blood picture is within normal limits

B- The findings are typical of hemolytic anemia

C- The findings are typical of vitamin B12 deficiency anemia

D- this case should generally treated by iron supplements



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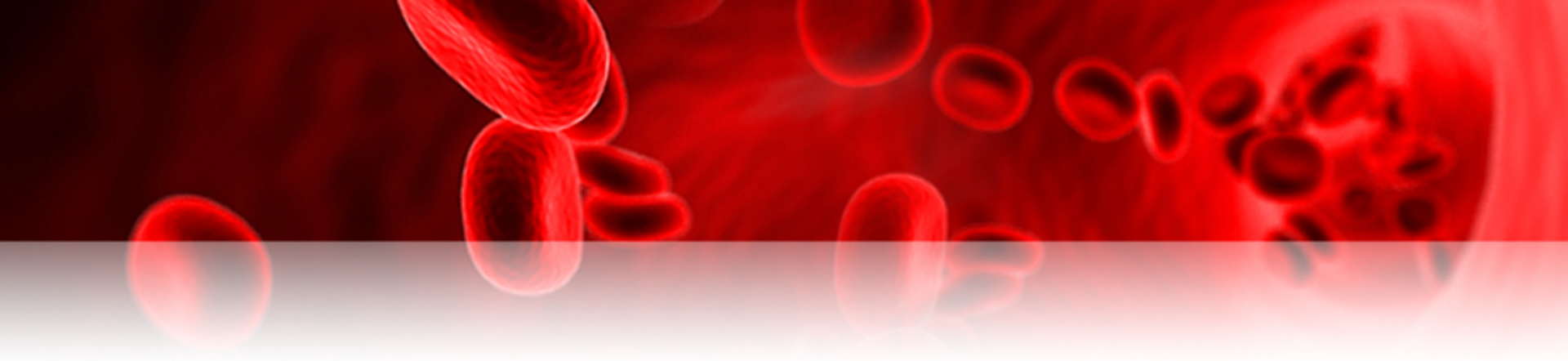
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Thank you